

CH32H417 Universal High-speed Interface (UHSIF) Development

Datasheet

Version: V1.0

<https://wch-ic.com>

1. Overview

The chip incorporates a set of universal high-speed interfaces (UHSIF), which can transmit clock up to 125MHz, support data widths of 8-, 16-, 24-, or 32-bit, with a theoretical maximum data rate of 500Mbytes/s.

2. Features

- Support master and slave modes
- Support 8-, 16-, 24-, and 32-bit data bus widths
- Configurable clock frequency, up to 125MHz
- Configurable clock sampling edge and data output edge
- Flexible mapping of address and control pins

3. Pin Information

3.1. Pin Definitions

The UHSIF peripheral supports up to 49 pins, including the UHSIF_CLK pin and the configurable interface pins UHSIF_PORT[47:0].

The interface pins consist of 2 address pins, 14 control pins, and 32 data pins.

In standard applications, a 32-bit data width requires 44 pins; for 24-bit, 16-bit, and 8-bit data widths, simply reduce the number of data pins accordingly. Refer to Table 3-1 for pin definitions.

If not all transmission lines (Line) are used, the number of SEL pins can be reduced accordingly. For example, if only Line0 and Line1 are used, only the SEL0 pin is required. Specifically, if only Line0 is used, no SEL pin is required. For information on transmission lines, refer to Section 4.1 Transmission Lines and Buffers.

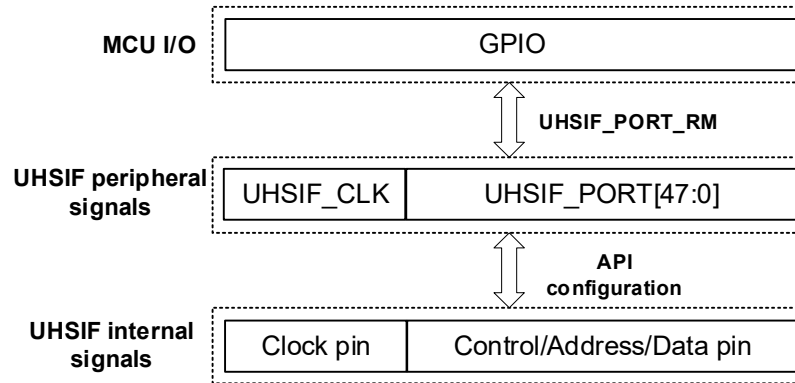
Table 3-1 Pin definitions

Pin name	Pin type	Pin direction		Pin function
		Host	Slave	
CS#	Control	O	I	Chip selection signal, active low.
WR#	Control	O	I	Write enable signal, active low.
RD#	Control	O	I	Read enable signal, active low.
OE#	Control	O	I	Slave output enable signal, active low.
EOP#	Control	O	I	Write end or write zero length packet signal, active low.
SEL[1:0]	Address	O	I	Address bus for internal transmission line selection.
DATA[31:0]	Data	I/O	I/O	32-bit data bus.
PCLK	Clock	O	I	Clock signal.
RDNE	Control	I	O	Non-null signal in slave buffer.
WRNF	Control	I	O	Non-full signal in slave buffer.

AF#	Control	I	O	Almost Full signal in slave buffer.
AE#	Control	I	O	Almost Empty signal in slave buffer.

3.2. Pin Mapping Description

Figure 3-1 UHSIF signal path



The UHSIF clock pin (PCLK) is permanently connected to UHSIF_CLK, while the remaining UHSIF pins can be mapped to UHSIF_PORT[47:0]. DATA[31:0] can be mapped to UHSIF_PORT[31:0] or UHSIF_PORT[47:16]; in either mapping scheme, the address and control pins can be flexibly configured on the remaining lines.

The UHSIF library provides a set of default mappings, as shown in Table 3-2. For other mappings, please contact technical support.

Table 3-2 Pin default mapping configuration

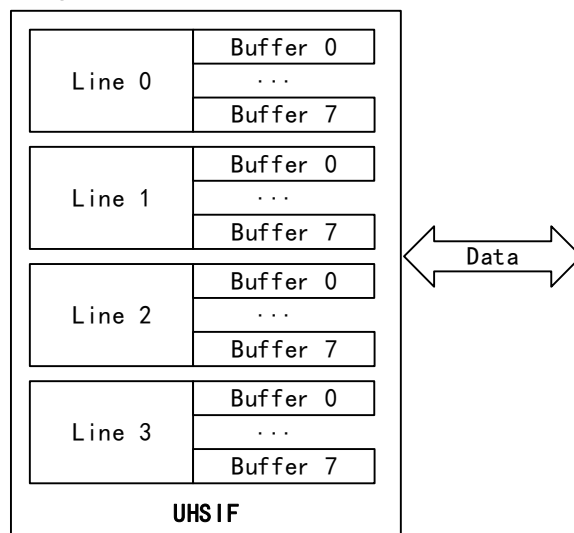
UHSIF_PORT	Pin function	UHSIF_PORT	Pin function	UHSIF_PORT	Pin function
UHSIF_PORT0	NC	UHSIF_PORT16	DATA[0]	UHSIF_PORT32	DATA[16]
UHSIF_PORT1	NC	UHSIF_PORT17	DATA[1]	UHSIF_PORT33	DATA[17]
UHSIF_PORT2	NC	UHSIF_PORT18	DATA[2]	UHSIF_PORT34	DATA[18]
UHSIF_PORT3	SEL[0]	UHSIF_PORT19	DATA[3]	UHSIF_PORT35	DATA[19]
UHSIF_PORT4	SEL[1]	UHSIF_PORT20	DATA[4]	UHSIF_PORT36	DATA[20]
UHSIF_PORT5	AF#	UHSIF_PORT21	DATA[5]	UHSIF_PORT37	DATA[21]
UHSIF_PORT6	NC	UHSIF_PORT22	DATA[6]	UHSIF_PORT38	DATA[22]
UHSIF_PORT7	AE#	UHSIF_PORT23	DATA[7]	UHSIF_PORT39	DATA[23]
UHSIF_PORT8	EOP#	UHSIF_PORT24	DATA[8]	UHSIF_PORT40	DATA[24]
UHSIF_PORT9	WRNF	UHSIF_PORT25	DATA[9]	UHSIF_PORT41	DATA[25]
UHSIF_PORT10	NC	UHSIF_PORT26	DATA[10]	UHSIF_PORT42	DATA[26]
UHSIF_PORT11	RDNE	UHSIF_PORT27	DATA[11]	UHSIF_PORT43	DATA[27]
UHSIF_PORT12	RD#	UHSIF_PORT28	DATA[12]	UHSIF_PORT44	DATA[28]
UHSIF_PORT13	OE#	UHSIF_PORT29	DATA[13]	UHSIF_PORT45	DATA[29]
UHSIF_PORT14	WR#	UHSIF_PORT30	DATA[14]	UHSIF_PORT46	DATA[30]
UHSIF_PORT15	CS#	UHSIF_PORT31	DATA[15]	UHSIF_PORT47	DATA[31]

The mapping of UHSIF peripheral signals to MCU I/O ports can be configured via the UHSIF_PORT_RM register. For details, refer to the "UHSIF Pin Functions" section in the *CH32H417DS0.PDF*.

4. Functional Description

4.1. Transmission Lines and Buffers

Figure 4-1 UHSIF transmission line and buffer



Transmission lines (Lines) are the 4 data channels used for high-speed data transfer within the UHSIF. They are selected via the external address lines SEL[1:0]. When the address lines switch, the transmission lines can switch without delay to continue data transmission and reception.

Each transmission line contains 8 buffers. By configuring these buffers, the UHSIF can implement up to 32 independent data paths. Each transmission line can notify the CPU of the status of its internal buffers via interrupts.

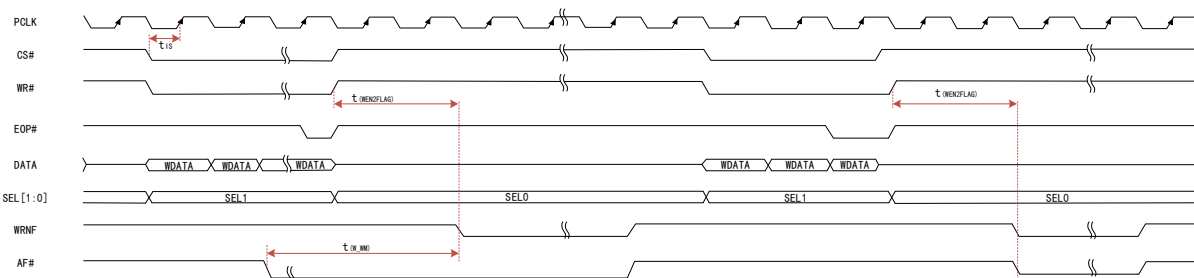
4.2. Communication Timing Description

4.2.1. Master Write Slave

When the UHSIF master writes data to the slave, it is divided into 2 processes: writing non-zero length packets and zero length packets.

- The master writes non-zero length packets to the slave

Figure 4-2 Timing of master writing non-zero length packet to slave

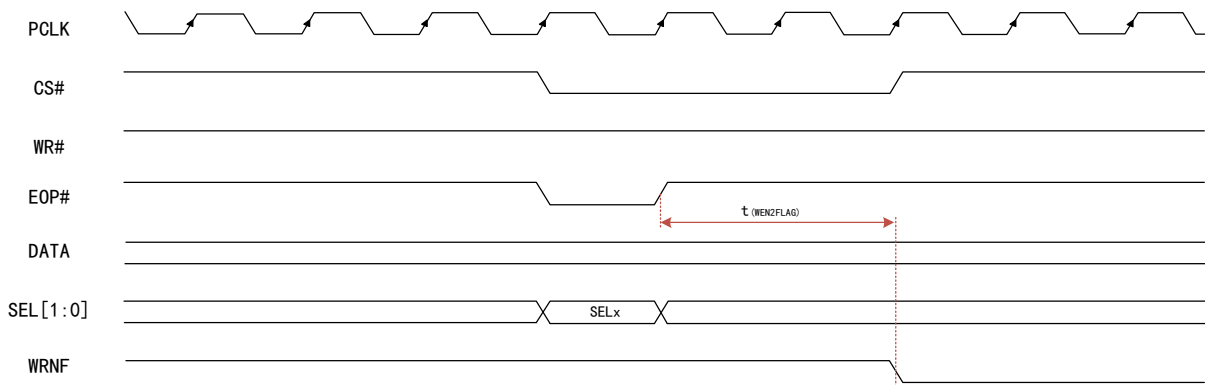


When the WRNF signal is valid, the UHSIF master can initiate a data write operation to the slave. According to the length of data written by the master, it can be divided into the following 2 situations:

1. When the length of data sent by the master is less than or equal to the length of data buffer received by the slave:
 - (1) The master drives the SEL1 and SEL0 signals to select the appropriate transmission line;
 - (2) The master drives the CS# and WR# signals and sends data to the data bus;
 - (3) After writing the data, the master drives the WR# signal to a high level; the slave invalidates the WRNF signal after 2 clock cycles.
2. When the length of data sent by the master is greater than the length of data buffer received by the slave:
 - (1) The master drives the SEL1 and SEL0 signals to select the appropriate transmission line;
 - (2) The master drives the CS# and WR# signals and sends data to the data bus;
 - (3) When the slave detects that the receive buffer is about to become full, it drives the AF# signal to a low level to notify the master. The master then sets WR# to a high level based on the watermark value to complete the write process, and the slave sets WRNF to an invalid state after 2 clock cycles.

● Zero-length packet

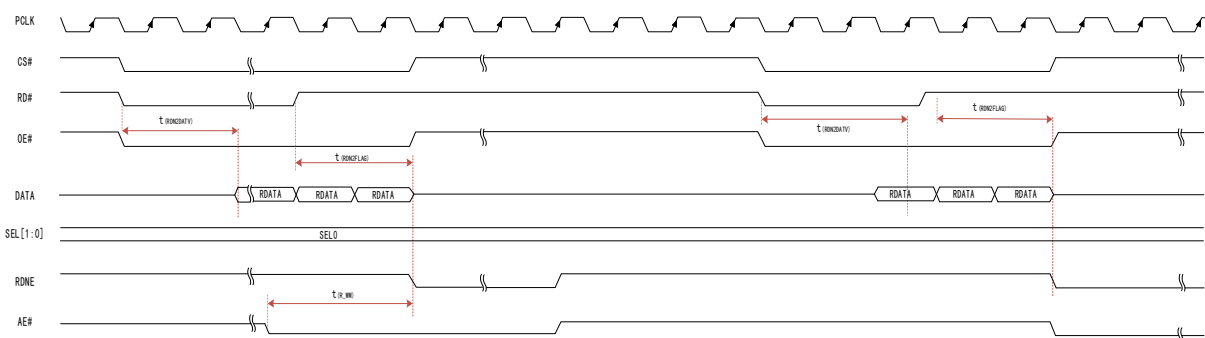
Figure 4-3 Timing of master writing zero-length packets to slave



1. The host drives the SEL1 and SEL0 signals to select the appropriate transmission line (Line);
2. The host pulls the EOP# and CS# signals low to indicate that the current transmission is a zero-length packet;
3. After transmission is complete, the host pulls the EOP# and CS# signals high.

4.2.2. Master Read Slave

Figure 4-4 Timing of master reading zero-length packets to slave



When RDNE is active, the master can initiate a data read. Depending on the length of data the master can receive, there are 2 scenarios:

1. When the length of data the master can receive is less than or equal to the length of data sent by the slave:

- a The master drives the SEL1 and SEL0 signals to select the appropriate data line;
 - b The master drives the OE# and CS# signals, and the data bus is then controlled by the slave;
 - c The master drives RD#, and the slave returns data to the master after 2 clock cycles;
 - d After reading the data, the master sets RD#, CS#, and OE# to high, ending the read process; the slave sets RDNE to invalid 2 clock cycles after the RD# high level expires.
2. When the length of data the master can receive is greater than the length of data sent by the slave:
- a The master drives the SEL1 and SEL0 signals to select the appropriate data line;
 - b The master pulls the OE# and CS# signals low, handing control of the data bus over to the slave;
 - c The master sets RD# to a low level; the slave returns data to the host after two clock cycles;
 - d When the slave detects that the host is about to finish reading the transmit buffer, it sets AE# to a low level to notify the master. The master then sets RD#, CS#, and OE# to high levels based on the watermark value, complete the read process. The slave sets RDNE to an invalid state 2 clock cycles after the RD# high level expires.

Table 4-2 Master read/write slave parameters

Parameter	Description	Fixed value	Unit
$t_{(WEN2FLAG)}$	The time from when the WR#/EOP# signal is pulled high until the WRNF signal going low when write slave is completed	2	cycle
$t_{(W_WM)}$	The time at which the AF# is lowered in advance when the buffer is about to fill up	(1)	cycle
$t_{(RDN2DATV)}$	The time from when the RD# signal is pulled low until valid data appears on the data bus when read slave	2	cycle
$t_{(RDN2FLAG)}$	The time from when the RD# signal is pulled high until slave RDNE going low when read slave is completed	2	cycle
$t_{(R_WM)}$	The time at which AE# is lowered in advance when the buffer is about to run out	(1)	cycle

Note: 1. Defined by the "water_mark_value" parameter in the UHSIF_Line_Cfg API.

5. UHSIF Library Function Description

5.1. Host Configuration Process

1. Use UHSIF_GPIO_Init to configure the host mode, pin mapping, clock frequency, and data bus width;

```
UHSIF_GPIO_Init( DEF_UHSIF_MASTER_MODE,
                  DEF_UHSIF_PINREMAP2,
                  DEF_UHSIF_CLKREMAP1,
                  RCC_UHSIFDIV_DIV4,
                  DEF_UHSIF_DATA_BIT32 );
```

2. Configure related registers;

```
UHSIF_Cfg( );
```

3. Configure the transmission line, the number of buffers, the send and receive buffer addresses, and the watermark value;

```
UHSIF_Line_Cfg( DEF_LINE0,
                 DEF_LINE_DIR_IN,
                 DEF_BUFF_COUNT_8,
                 USBSS_EP1_Tx_Buf,
                 0 );
UHSIF_Line_Cfg( DEF_LINE1,
                 DEF_LINE_DIR_OUT,
                 DEF_BUFF_COUNT_8,
                 USBSS_EP1_Rx_Buf,
                 0 );
```

4. Start the transmission; by default, the system will receive one packet of data after the transmission begins;

```
UHSIF_Start( ENABLE );
```

5. To continue sending or receiving data, use the UHSIF_Trans_Cfg function. In this function, DEF_LINE0 enables data reception on transmission line 0 until the UHSIF_RD_COMM.death buffer is received; when receiving data, simply set the trans_size parameter to 0.

```
UHSIF_Trans_Cfg( DEF_LINE0, UHSIF_RD_COMM.death, 0 );
```

5.2. Slave Configuration Process

1. Use UHSIF_GPIO_Init to configure the slave mode, pin mapping, clock frequency, and data bus width. Select either FPGA mode or SOC mode for the slave mode depending on your application;

```
UHSIF_GPIO_Init( DEF_UHSIF_SLAVE_SOC_MODE,
                  DEF_UHSIF_PINREMAP2,
                  DEF_UHSIF_CLKREMAP1,
                  0,
                  DEF_UHSIF_DATA_BIT32 );
```

2. Configure related registers;

```
UHSIF_Cfg();
```

3. Configure the transmission line, the number of buffers, the send and receive buffer addresses, and the watermark value;

```
UHSIF_Line_Cfg( DEF_LINE0,
                 DEF_LINE_DIR_IN,
                 DEF_BUFF_COUNT_8,
                 USBSS_EP1_Tx_Buf,
                 4094 );
UHSIF_Line_Cfg( DEF_LINE1,
                 DEF_LINE_DIR_OUT,
                 DEF_BUFF_COUNT_8,
                 USBSS_EP1_Rx_Buf,
                 4 );
```

4. Start the transmission; by default, the system will receive one packet of data after the transmission begins;

```
UHSIF_Start( ENABLE );
```

5. To continue sending or receiving data, use the UHSIF_Trans_Cfg function, where DEF_LINE0 enables data reception on transmission line 0. This will receive data until the UHSIF_RD_COMM.death buffer is reached; when receiving, simply set the trans_size parameter to 0.

```
UHSIF_Trans_Cfg( DEF_LINE0, UHSIF_RD_COMM.death, 0 );
```

5.3. API Description

5.3.1. UHSIF_Get_Ver

Get the current UHSIF library version number.

Interface definition:

```
uint8_t UHSIF_Get_Ver( void );
```

Parameter	Description
Return	Current UHSIF library version number

5.3.2. UHSIF_Cfg

Initialize the UHSIF registers. This function must be called after the UHSIF GPIO has been initialized and before the transmission lines are initialized.

Interface definition:

```
void UHSIF_Cfg( void );
```

Parameter	Description
None	Initialize UHSIF related registers

5.3.3. UHSIF_GPIO_Init

UHSIF pin initialization is used to configure master and slave modes, as well as to configure pin remapping. UHSIF modes include slave FPGA mode, slave SOC mode, and host mode. When the master is an FPGA, the slave must use FPGA mode; the slave SOC mode is only applicable when connecting to a host of the same type.

The maximum clock frequency for UHSIF is 125MHz, derived from the SYSCCLK clock. When UHSIF is in master mode, the clock division factor can be configured via `clk_div`.

Interface Definition:

```
uint8_t UHSIF_GPIO_Init( uint8_t mode_select,
                          uint8_t uhsif_port_rm,
                          uint8_t uhsif_clk_rmm,
                          uint8_t clk_div,
                          uint8_t width_bit );
```

Parameter	Description
<code>mode_select</code>	Select to configure master mode or slave mode.
<code>uhsif_port_rm</code>	Select the remapping pin of UHSIF control line and data line.
<code>uhsif_clk_rmm</code>	Select the remap pin of the UHSIF clock line.
<code>clk_div</code>	Clock frequency division coefficient in UHSIF master mode
<code>width_bit</code>	Select the data bus width as 8/16/24/32-bit.
Return	Whether Pin initialization is successful

5.3.4. UHSIF_Line_Cfg

UHSIF transmission line initialization is used to configure the transmission line number, transmission direction, number of buffers used by the transmission line, and the addresses of the send and receive buffers for the current transmission line.

Since UHSIF has a total of 4 transmission lines, the `line_num` value can range from 0 to 3;

This function allows for the configuration of multiple transmission lines for data transmission and reception; however, each transmission line can only be configured for a single direction.

Each transmission line supports up to 8 buffers, so `buff_count` can be set to a value between 1 and 8.

`water_mark_value` specifies the number of cycles in advance by which the signal should be pulled low upon completion of transmission. For example, if `DEF_BUFF_SIZE` is set to 16K and you want to pull the signal low 2 cycles in advance, the value of `water_mark_value` is $16384/16 - 2 = 4094$.

Interface Definition:

```
uint8_t UHSIF_Line_Cfg(uint8_t line_num,
                       uint8_t line_dir,
                       uint8_t buff_count,
                       uint8_t *pbuf,
                       uint16_t water_mark_value);
```

Parameter	Description
line_num	Select the transmission line number to initialize
line_dir	Current transmission line direction configuration
buff_count	Current transmission line BUFF number configuration
*pbuf	Address of buffer used to transmit and receive data on current transmission line
water_mark_value	Indicates the number of words remaining to be transmitted in the buffer when transmitted and the number of words to be received in the buffer when received
Return	Whether transmission line configuration is successful

5.3.5. UHSIF_Start

UHSIF activated. After the function is configured, the first buff of the configured receiving transmission line enables data reception once.

Interface definition:

```
void UHSIF_Start( FunctionalState sta );
```

Parameter	Description
sta	Select whether to enable parallel port function

5.3.6. UHSIF_Linux_IN_Callback

The UHSIF transmission line receives an interrupt callback function; this interrupt is triggered when the transmission line receives a long or short packet. The 'rcv_count' value read represents the length of the data received during this transmission.

Interface definition:

```
void UHSIF_Linux_IN_Callback( uint32_t rcv_count );
```

Parameter	Description
rcv_count	Length of data received on the current transmission line

5.3.7. UHSIF_Linux_OUT_Callback

The UHSIF transmission line sends an interrupt callback function; this interrupt function is called once the transmission line has finished sending data.

Interface definition:

```
void UHSIF_Linux_OUT_Callback( void );
```

Parameter	Description
None	UHSIF transmission line send interrupt callback function

5.3.8. UHSIF_Trans_Cfg

Enable a single UHSIF data transmission or reception.

Since UHSIF has a total of 4 transmission lines, the line_num value can be set to 0–3;

Based on the buff_count value configured via the UHSIF_Line_Cfg function, buff_cur can be set to any value between 0 and buff_count – 1; it is incremented and reused in a loop during operation;

When configuring a transmission line to send data, the trans_size value represents the length of the data to be sent; when configuring a transmission line to receive data, this value is irrelevant and should be set to 0.

Interface definition:

```
uint8_t UHSIF_Trans_Cfg( uint8_t line_num,
                        uint8_t buff_cur,
                        uint32_t trans_size );
```

Parameter	Description
line_num	Select the transmission line number to use
buff_cur	Select the number of buffs to use for this transmission line
trans_size	Configure the length of data sent this time
Return	Whether the configuration parameter is valid

5.3.9. UHSIF_Reg_Change

The UHSIF has a total of 76 registers that can be modified; the offset address reg_addr ranges from 0 to 75, and value should be set to the value you wish to modify.

```
uint8_t UHSIF_Reg_Cfg(uint8_t mode,
                      uint8_t reg_addr,
                      uint32_t value );
```

Parameter	Description
mode	Master/Slave register select
reg_addr	Register address
value	Register value to be modified
Return	whether the operation is successful

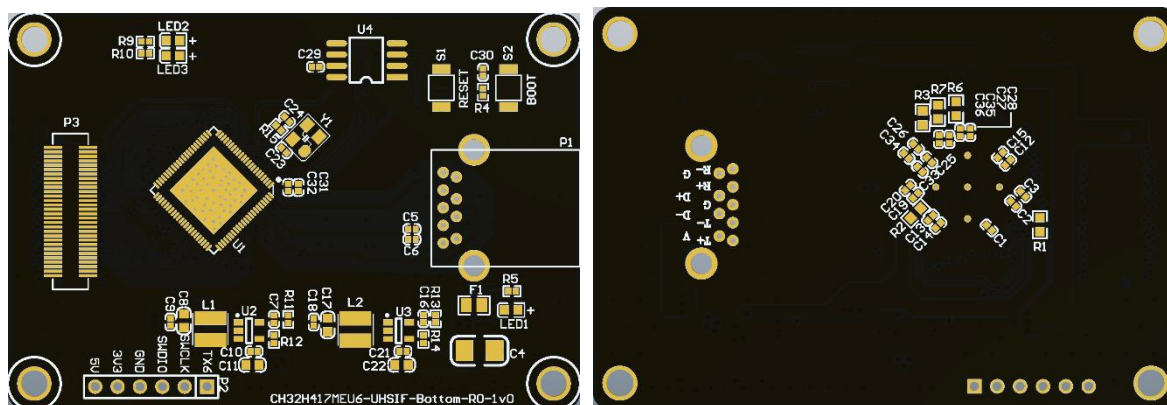
6. Introduction to Evaluation Board

CH32H417 UHSIF offers a variety of evaluation boards for use with this product.

6.1. CH32H417MEU6-UHSIF-Bottom-R0-1v0

CH32H417MEU6 UHSIF Evaluation Board base board, featuring an on-board debug interface, USB3.0 port, external Flash, and UHSIF interface. It is compatible with multiple evaluation board top plates.

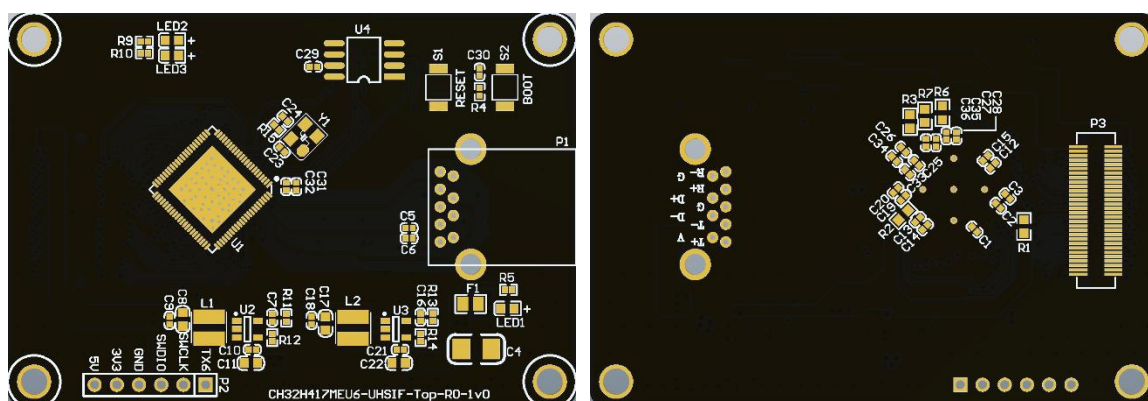
Figure 6-1 CH32H417MEU6-UHSIF-Bottom-R0-1v0 evaluation board



6.2. CH32H417MEU6-UHSIF-Top-R0-1v0

CH32H417MEU6 UHSIF Evaluation Board Top Panel, featuring an on-board debug interface, USB3.0 port, external Flash, and UHSIF interface. Must be used in conjunction with the CH32H417MEU6-UHSIF-Bottom-R0. Suitable for demonstrating UHSIF-related examples.

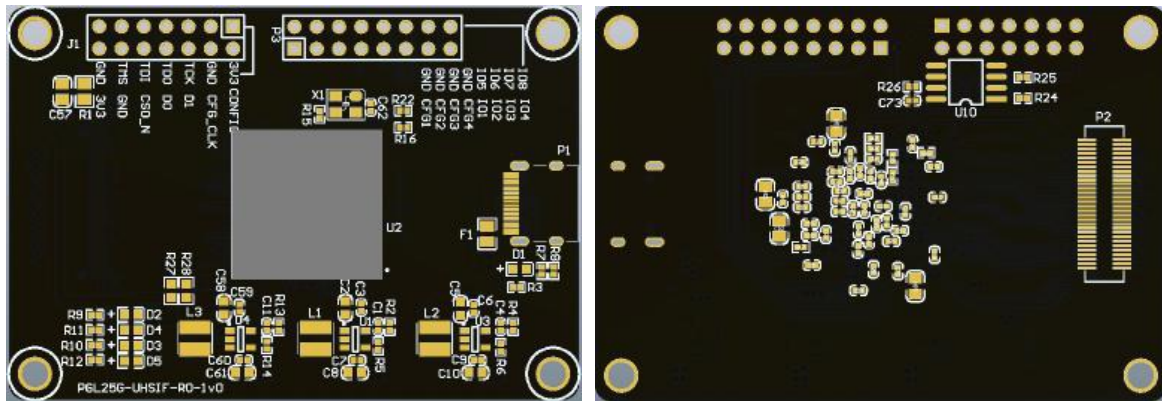
Figure 6-2 CH32H417MEU6-UHSIF-Top-R0-1v0 evaluation board



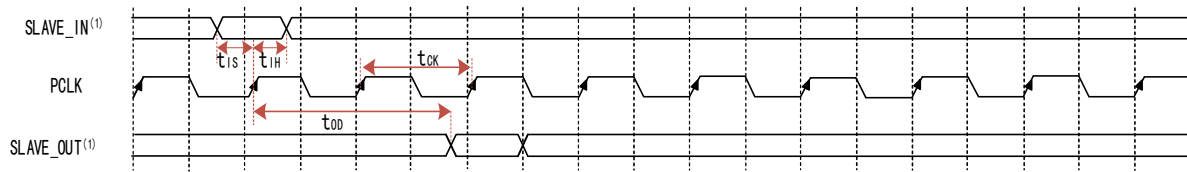
6.3. FPGA-UHSIF-Top-R0-1v0

UHSIF-compatible FPGA evaluation board, designed for use with the CH32H417MEU6-UHSIF-Bottom-R0. It can be used to demonstrate UHSIF-related examples.

Figure 6-3 FPGA-UHSIF-Top-R0-1v0 evaluation board



7. Electrical Characteristics



Note: 1. SLAVE_IN refers to the slave input, and SLAVE_OUT refers to the slave output.

Table 4-2 Slave electrical parameters

Symbol	Parameter description	Condition	Min.	Max.	Unit
f_{CK}/t_{CK}	Clock frequency in data transmission mode	$CL \leq 10pF$, $V_{DD18} = 1.8V$		125	MHz
t_C	Clock cycle		8		ns
t_{IS}	Settling time from the input signal to CLK		0.5		ns
t_{IH}	Holding time from slave input signal to CLK		1.5		ns
t_{OD}	Delay from slave output signal to CLK		6.2	8	ns